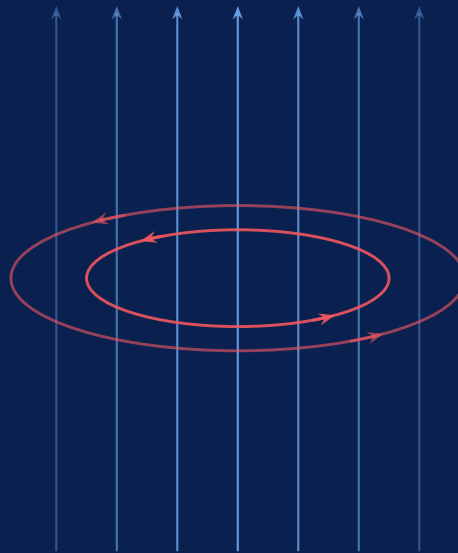


6



CHAPTER

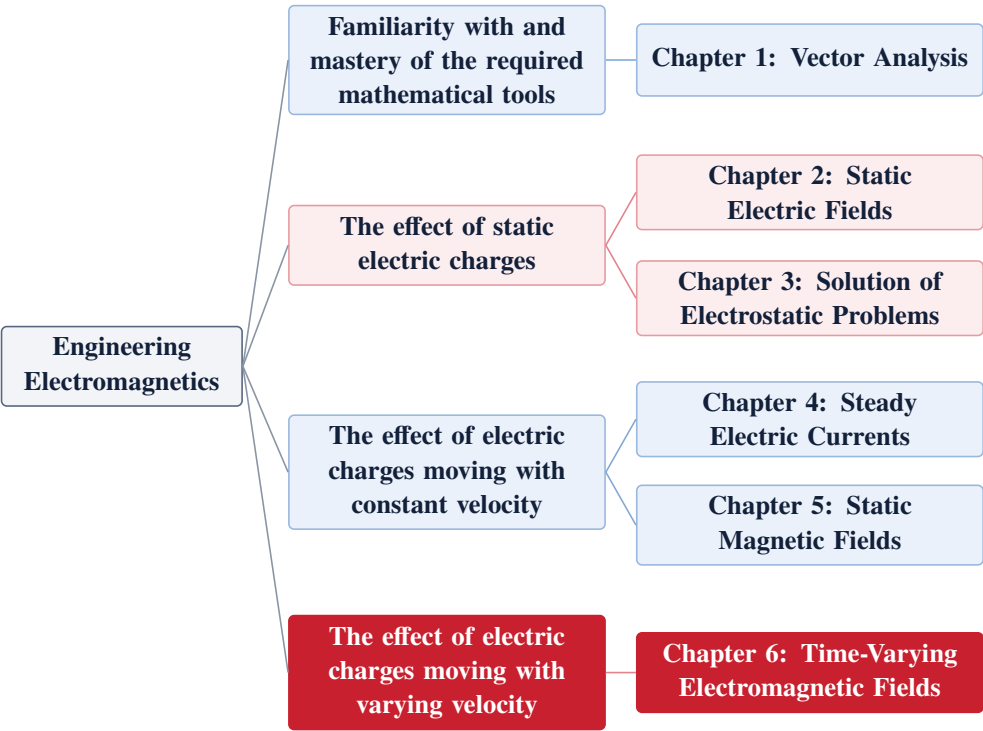
Time-Varying Electromagnetic Fields

میدان‌های الکترومغناطیسی متغیر با زمان

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6.1 Course Outline



6.2 Introduction

- The fundamental relations of electrostatics and magnetostatics

Fundamental Relations	Electrostatic Model	Magnetostatic Model
Governing equations	$\nabla \times \mathbf{E} = 0, \nabla \cdot \mathbf{D} = \rho$	$\nabla \cdot \mathbf{B} = 0, \nabla \times \mathbf{H} = \mathbf{J}$
Constitutive relations (linear and isotropic media)	$\mathbf{D} = \epsilon \mathbf{E}$	$\mathbf{H} = \frac{1}{\mu} \mathbf{B}$

- In the static (time-invariant) case, the electric field vectors and the magnetic field vectors form separate, independent pairs.
- We shall see that in the time-varying case a time-varying magnetic field produces an electric field, and vice versa

6.3 Faraday's Law of Electromagnetic Induction

- In 1820 **Hans Christian Ørsted** showed that a wire carrying an electric current deflects a compass needle.
 - The motion of the compass needle was caused by the magnetic force of the magnetic field produced by the electric current.
- After this discovery **Michael Faraday** reasoned that a magnetic field should, in the same way, have an electric effect and produce a current in a wire.
- After about ten years of experiments he made the discovery known as Faraday's law of electromagnetic induction.

≡ Definition | Faraday's law of electromagnetic induction

If the magnetic flux through a closed circuit changes with time, an electromotive force is induced in the circuit. This electromotive force drives a current.

- The polarity of the electromotive force, and hence the direction of the induced current, can be found from Lenz's law.

≡ Definition | Lenz's law

The current induced in a circuit always flows in the direction that opposes the change of the magnetic flux producing it.

- The mathematical statement of Faraday's law of electromagnetic induction

⚡ Important result

$$v_{emf} = -N \frac{d\Phi}{dt} = -N \frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{s}$$

- The definition of the electromotive force induced in a circuit with the closed contour C

$$v_{emf} \triangleq \oint_C \mathbf{E} \cdot d\boldsymbol{\ell}$$

- For a stationary circuit

$$v_{emf} = \oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} \quad \Rightarrow \quad \int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$$

- The last step uses Stokes's theorem.

⚡ Important result

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

- In the time-varying case the electric field is not conservative $\rightarrow \nabla \times \mathbf{E} \neq 0$
- In the time-varying case the scalar potential cannot be defined as $V = -\int \mathbf{E} \cdot d\ell$, because the potential difference between two points would depend on the path of integration and would not have a fixed value.
- At low frequencies $\rightarrow \nabla \times \mathbf{E} \simeq 0 \rightarrow$ the scalar potential can be defined (lumped-circuit theory)

🔗 Example 6-1

A circular loop of N turns of conducting wire lies in the xy -plane with its centre at the origin, in a magnetic field given by

$$\mathbf{B} = \mathbf{a}_z B_0 \cos\left(\frac{\pi r}{2b}\right) \sin(\omega t)$$

Here b is the radius of the loop and ω the angular frequency. Find the induced electromotive force.

► Solution.

$$\begin{aligned} \Phi &= \int_S \mathbf{B} \cdot d\mathbf{s} = \int_0^{2\pi} \int_0^b \left[B_0 \cos\left(\frac{\pi r}{2b}\right) \sin(\omega t) \mathbf{a}_z \right] \cdot (r dr d\phi \mathbf{a}_z) \\ &= \frac{8b^2}{\pi} \left(\frac{\pi}{2} - 1 \right) B_0 \sin(\omega t) \\ v &= -N \frac{d\Phi}{dt} = -\frac{8Nb^2}{\pi} \left(\frac{\pi}{2} - 1 \right) B_0 \omega \cos(\omega t) \end{aligned}$$

6.4 Displacement Current

- Ampère's circuital law for the static magnetic field

$$\nabla \times \mathbf{H} = \mathbf{J}$$

- James Clerk Maxwell doubted this equation, because

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0 = \nabla \cdot \mathbf{J}$$

- which is not consistent with the equation of continuity

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}$$

- To remove this inconsistency Maxwell modified the relation as follows

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0 = \nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} \quad \xrightarrow{\nabla \cdot \mathbf{D} = \rho} \quad \nabla \cdot (\nabla \times \mathbf{H}) = \nabla \cdot \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right)$$

⚡ Important result

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

- It can easily be shown that $\partial \mathbf{D} / \partial t$ has the dimension of a current density. It is called the **displacement current density**.
- The modified form of Ampère's circuital law states that a time-varying electric field produces a magnetic field even when there is no current.

6.5 Maxwell's Equations

- The equations governing time-varying electric and magnetic fields, or **Maxwell's equations**

Differential Form	Integral Form	Significance
$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi}{dt}$	Faraday's law
$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = I + \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}$	Ampère's circuital law
$\nabla \cdot \mathbf{D} = \rho$	$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$	Gauss's law
$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$	No isolated magnetic charge